

Joint Predictions and Counterfactuals under Symmetric Latent Information*

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Abstract

This paper studies joint prediction and counterfactual analysis for an observed game and a counterfactual game when the information structure is latent to the analyst but common across the two. The observed and counterfactual games are assumed to belong to a maintained symmetric class: payoffs are symmetric, private types are exchangeable, and additional signals satisfy a symmetric conditional-independence restriction. I construct a linked game that encodes fixed information across the two games. SBCE of the linked base game yields an exact characterization of the joint outcome distributions induced by symmetric Bayes Nash equilibrium as the common admissible information structure ranges over the class. The resulting fixed-information counterfactual prediction set is contained in the unrestricted BCE counterfactual prediction set of the same linked base game. In the two-firm entry environment of Bergemann et al. (2022), this containment is strict: for positive payoff shifters $z \in (0, 1)$, the SBCE producer-surplus prediction set is the singleton $\mathcal{W}^{\text{SBCE}}(z) = \{2\}$, whereas unrestricted BCE of the linked base game allows producer surplus to move both above and below 2. Thus these primitive restrictions can have counterfactual content beyond the fixed-information requirement itself.

Keywords: Bayes correlated equilibrium, Symmetric Bayes Correlated Equilibrium, fixed-information counterfactuals, symmetric games, latent information

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1 Introduction

Counterfactual analysis in games asks what behavior can arise in one environment when equilibrium behavior is observed in another and the underlying information structure is held fixed. The difficulty is that the object to be held fixed is itself latent to the analyst. With a known information structure, the problem would reduce to counterfactual equilibrium analysis under specified primitives. Here that structure is unobserved and may encode

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rich private information and beliefs. The challenge is therefore to derive counterfactual implications without specifying the information structure.

Bayes correlated equilibrium (BCE) provides a reduced-form approach to games in which the information structure is latent to the analyst. In this approach, the payoff and action primitives, together with the prior over payoff-relevant uncertainty, form the base game; an expansion of the base game specifies the additional information that players receive. Bergemann and Morris (2016) show that the set of distributions induced by Bayes Nash equilibria over all information expansions of a static base game coincides with the set induced by BCE of that base game. Building on this idea, Bergemann et al. (2022) study fixed-information counterfactuals by linking an observed game and a counterfactual game under a common latent information structure and then characterizing the associated linked game through BCE. Their approach isolates the predictive content of fixed latent information. In the symmetric environments studied here, counterfactual predictions are shaped not only by the requirement that latent information be held fixed, but also by the primitive restrictions that define the maintained class of games. The question is how those class restrictions should enter fixed-information counterfactual analysis.

I study fixed-information counterfactuals within a symmetric class of static games with latent information. In both the observed and counterfactual games, payoffs are symmetric across players, private types are exchangeably distributed, and additional private signals satisfy a symmetric conditional-independence restriction. These are primitive restrictions on the environments the analyst is willing to maintain, not restrictions imposed directly on outcome distributions. This class allows noisy private information about opponents' types but rules out arbitrary cross-player correlation in the additional-signal layer. The issue is how to carry these primitive restrictions into fixed-information counterfactual analysis.

This paper shows that fixed-information counterfactual predictions depend on the maintained primitive class, not only on the requirement that information be held fixed. In the symmetric latent-information framework studied here, the restrictions are imposed before the model is reduced to outcome distributions: payoff symmetry and exchangeability discipline the payoff-relevant environment, while symmetric equilibrium behavior and symmetric conditional independence discipline how outcome distributions can be generated. Because some of these restrictions live in the latent additional-signal layer, they cannot be imposed

as simple ex post restrictions on observed or counterfactual outcome distributions.

I characterize the linked counterfactual problem through SBCE of the linked base game. The linked-game outcome distributions generated by symmetric Bayes Nash equilibrium over admissible symmetric expansions coincide with the distributions induced by SBCE of the linked base game. Given the observed marginal, the admissible counterfactual distributions are obtained by projecting those SBCE distributions that match it.

The comparison with unrestricted BCE is made holding fixed the same linked base game and the same observed marginal. Thus both constructions inherit the payoff and type-state restrictions encoded in the linked base game, including payoff symmetry and exchangeability of private types. The difference is that SBCE of the linked base game also reflects the maintained restrictions not encoded by the linked base game itself: symmetric equilibrium behavior and symmetric conditional independence in the latent additional-signal layer. Hence the SBCE counterfactual set is contained in the unrestricted BCE counterfactual set.

This difference is not only formal. The two-firm entry environment of Bergemann et al. (2022) shows that the restrictions of the symmetric class can matter for a welfare-relevant comparative static. In that environment, a counterfactual payoff shifter z raises the payoff from entry in both monopoly and duopoly states. For $z \in (0, 1)$, unrestricted BCE of the same linked base game permits producer-surplus predictions both above and below 2, the level generated by the symmetric mixed-strategy analysis. Under SBCE of the linked base game, by contrast, the producer-surplus prediction set is the singleton

$$\mathcal{W}^{\text{SBCE}}(z) = \{2\}, \quad z \in (0, 1).$$

The neutrality is not because the payoff shifter is irrelevant. The higher payoff from entry induces additional high-cost entry, and the resulting adjustment in market structure exactly offsets the direct payoff increase. Thus the primitive restrictions defining the symmetric class can have counterfactual content beyond the fixed-information requirement itself: they can change the welfare-relevant comparative static.

This paper builds on the literature that uses BCE to study games with latent information and fixed-information counterfactuals. This line of work was initiated by Bergemann and Morris (2013, 2016) and extended to fixed-information counterfactuals by Berge-

mann et al. (2022).¹ Relative to Bergemann et al. (2022), the analysis studies fixed-information counterfactuals under a maintained symmetric latent-information class. The single-environment prediction problem for this class is studied in Lu (2026). The present paper addresses a distinct linked problem: how the same primitive restrictions constrain joint predictions and counterfactual distributions when one common latent information structure must rationalize both the observed and counterfactual games. These joint restrictions are absent from the single-game setting and go beyond the Bayes-consistency and obedience restrictions imposed by unrestricted BCE of the linked base game.

The paper is also related to the broader literature on correlated equilibrium in incomplete information environments, including Forges (1993), Forges (2006), and Liu (2015). That literature studies which informational restrictions should be imposed on correlated-equilibrium objects in games with incomplete information.² The route taken here is different. I take the symmetric latent-information class as the primitive model and derive the exact linked-game characterization implied by that class. Thus SBCE of the linked base game is not a refinement imposed on unrestricted BCE; it is the reduced-form object that characterizes the linked-game distributions generated by the maintained class. The paper is therefore not about equilibrium selection in symmetric games more generally.

The paper is theoretical in scope, but the characterization has immediate counterfactual content. It shows how primitive restrictions on latent information enter fixed-information predictions through SBCE of the linked base game. The entry environment illustrates this content by showing that these restrictions can turn an unrestricted BCE producer-surplus interval into a singleton SBCE prediction.

The rest of the paper is organized as follows. Section 2 introduces the component games,

¹BCE-based approaches have also been used in subsequent work; see, for example, Magnolfi and Roncoroni (2023) on discrete games, Gualdani and Sinha (2024) on discrete choice, and Syrgkanis et al. (2021) on auctions. Related dynamic and extensive-form extensions include Makris and Renou (2023), Hara et al. (2025), and Doval and Ely (2020).

²Bayesian persuasion and information design study the choice of information structures to influence beliefs, behavior, and equilibrium outcomes (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019). BCE provides a natural reduced-form language for such problems. The present paper does not study optimal signal design or equilibrium selection under a designer’s objective; it studies equilibrium characterization under latent information for a symmetric primitive class.

the common information structure, and the maintained symmetric latent-information class. Section 3 constructs the linked game, defines the SBNE-induced distributions in that game, and verifies closure of the maintained class under linking. Section 4 defines SBCE of the linked base game and derives the joint-prediction characterization. Section 5 derives the fixed-information counterfactual characterization. Section 6 compares the resulting counterfactual set with the unrestricted BCE set for the same linked base game. Section 7 studies the two-firm entry environment. Section 8 concludes. Appendix A contains the formal equilibrium definition and proofs; Appendix B gives a primitive representation of the SBCE feasible set for the linked base game.

2 Component games and maintained class

This section introduces the observed game, the counterfactual game, and the common information structure that is held fixed across them. It also defines the symmetric latent-information class maintained in the analysis and records the primitive restrictions that the linked construction must preserve.

2.1 Component games and common information

Consider two static games, indexed by $\ell \in \{0, 1\}$. Game $\ell = 0$ is the observed game, and game $\ell = 1$ is the counterfactual game. The two games share a common latent-information environment but may differ in their action spaces and payoff functions.

There is a finite set of players, indexed by $i = 1, \dots, N$, and a finite set of public states Θ . The public state $\theta \in \Theta$ is observed before actions are chosen and is therefore common knowledge. Let T be a finite private type space, and let each S_i be a measurable signal space. Each player i privately observes a pair

$$(t_i, s_i) \in T \times S_i,$$

where $t_i \in T$ is player i 's private type and $s_i \in S_i$ is an additional private signal. The type t_i may include payoff-relevant components as well as components that affect the distribution of additional signals. After introducing these objects, I refer to θ , t_i , and s_i simply as the state, type, and signal, respectively, when no confusion can arise. Let

$$t = (t_1, \dots, t_N) \in T^N, \quad s = (s_1, \dots, s_N) \in \mathcal{S} := \prod_{i=1}^N S_i.$$

The additional signal s_i may contain information about opponents' private types.

The common-prior information structure is

$$I = (T, (S_i)_{i=1}^N, \pi), \quad \pi \in \Delta(T^N \times \mathcal{S} \times \Theta).$$

Let $\pi^{T\Theta}$ denote the marginal distribution of (t, θ) induced by the common prior π . This marginal distribution is common across the observed and counterfactual games.

The two component games may differ in their action spaces and payoff functions. For each $\ell \in \{0, 1\}$, let A^ℓ denote the finite action set in component game ℓ , and let $\mathcal{A}^\ell := (A^\ell)^N$ denote the corresponding action-profile space. Payoffs in component game ℓ are given by

$$u_i^\ell : \mathcal{A}^\ell \times T^N \times \Theta \rightarrow \mathbb{R}.$$

Let $\mathcal{G}^\ell := ((u_i^\ell)_{i=1}^N, A^\ell, \Theta)$ denote the payoff and action primitives of component game ℓ , and let

$$\mathcal{E}^\ell := (\mathcal{G}^\ell, \pi^{T\Theta})$$

denote the corresponding payoff-relevant environment.

Thus the fixed-information requirement is that the two component games share the same information structure I , and hence the same marginal $\pi^{T\Theta}$ over private type profiles and public states. The payoff-relevant environments $\mathcal{E}^0$ and $\mathcal{E}^1$ are fixed in the analysis and are treated as known to the analyst.³ The latent part of the model is the additional-signal layer: the signal spaces $(S_i)_{i=1}^N$ and the conditional distribution of s given (t, θ) .

2.2 Maintained symmetric latent-information class

This subsection defines the symmetric latent-information class maintained in the analysis. Both the observed game and the counterfactual game are assumed to satisfy these restrictions when paired with the common information structure I .

For the additional-signal restriction, each S_i is interpreted as an opponent-indexed signal space. The spaces $(S_i)_{i=1}^N$ are identified through the natural relabelings of player identities;

³This is the analogue of the full-observation case in Bergemann et al. (2022). Weaker observational restrictions, such as restrictions on action marginals or moments, can be handled by replacing the observed-marginal matching constraint below with the corresponding feasibility restriction. I focus on the known- $\pi^{T\Theta}$ case to isolate the role of the maintained symmetric class.

these relabelings are suppressed when writing a common signal kernel. A permutation φ_{-i} of player i 's opponents acts on S_i , and on its measurable subsets, by relabeling the opponent-indexed coordinates.

The conditional law of the additional-signal profile given (t, θ) is used only on the support of $\pi^{T\Theta}$. Since $T^N \times \Theta$ is finite, for any (t, θ) with $\pi^{T\Theta}(t, \theta) > 0$, write $\pi(\cdot \mid t, \theta)$ for the conditional law of s given (t, θ) . No restriction is imposed on conditional versions off this support.

Definition 1 (Maintained symmetric latent-information class). *A single game $(\mathcal{G}, I)$, with common finite action set A and action-profile space $\mathcal{A} = A^N$, belongs to the maintained symmetric latent-information class if the following conditions hold:*

(i) **Symmetric payoffs.** *For every permutation ϕ of $\{1, \dots, N\}$, define*

$$\phi(a) := (a_{\phi(1)}, \dots, a_{\phi(N)}), \quad \phi(t) := (t_{\phi(1)}, \dots, t_{\phi(N)}).$$

Then, for every player i , every $a \in \mathcal{A}$, every $t \in T^N$, and every $\theta \in \Theta$,

$$u_i(a, t, \theta) = u_{\phi(i)}(\phi(a), \phi(t), \theta).$$

(ii) **Exchangeable private types.**

For every permutation ϕ of $\{1, \dots, N\}$ and every $(t, \theta) \in T^N \times \Theta$,

$$\pi^{T\Theta}(\phi(t), \theta) = \pi^{T\Theta}(t, \theta).$$

Thus the joint distribution of private types and the public state is invariant under relabeling of players.

(iii) **Symmetric conditionally independent additional private signals.** *There exists a common stochastic kernel*

$$\rho(\cdot \mid t_i, t_{-i}, \theta) \in \Delta(S_i)$$

such that, for every $(t, \theta) \in T^N \times \Theta$ with $\pi^{T\Theta}(t, \theta) > 0$, and every measurable rectangle

$$B = \prod_{i=1}^N B_i \subseteq \prod_{i=1}^N S_i,$$

one has

$$\pi(B \mid t, \theta) = \prod_{i=1}^N \rho(B_i \mid t_i, t_{-i}, \theta).$$

Moreover, for every player i , every type $t_i \in T$, every opponents' type profile $t_{-i} \in T^{N-1}$, every public state $\theta \in \Theta$ with $\pi^{T\Theta}(t_i, t_{-i}, \theta) > 0$, every permutation φ_{-i} of player i 's opponents, and every measurable set $B_i \subseteq S_i$,

$$\rho(B_i \mid t_i, t_{-i}, \theta) = \rho(\varphi_{-i}(B_i) \mid t_i, \varphi_{-i}(t_{-i}), \theta).$$

The restrictions in Definition 1 are primitive restrictions on the environments the analyst is willing to maintain. The private type profile t remains explicit in the payoff-relevant environment, and it also indexes the latent additional-signal technology through the kernel in condition (iii). Thus t_i may contain components that affect payoffs directly and components that affect the distribution of additional signals. By contrast, the additional signal profile s may affect beliefs but does not enter payoffs directly.

The conditional-independence condition restricts the additional-signal layer on the support of $\pi^{T\Theta}$: conditional on any (t, θ) with $\pi^{T\Theta}(t, \theta) > 0$, players' additional signals are independent across players and are drawn from a common opponent-symmetric signal kernel. The class therefore allows noisy private information about opponents' types, but rules out arbitrary cross-player correlation in the additional-signal profile conditional on the type-state profile. Symmetry is an ex ante restriction on the signal technology, not a requirement that all players have identical realized informational precision. Because the kernel may depend on the type profile through (t_i, t_{-i}, θ) , realized private types may generate heterogeneous informational advantages ex post. In this sense, the class nests independent private information, complete information, and intermediate noisy-signal environments in which realized private types may affect the precision of signals about opponents' types.

3 The linked game under fixed information

This section constructs the linked game associated with the two component games introduced above. The linked game replaces each player's action by a pair of component actions, one for the observed game and one for the counterfactual game, chosen under the same realization of the common information structure.

Because the common information structure is latent to the analyst, the linked problem separates a known payoff-relevant part from a latent additional-signal layer. The known payoff-relevant part is the base-game object in the sense of the BCE literature. The latent part is assumed to satisfy the informational restrictions of the maintained symmetric class. After defining the linked game, I introduce the distributions induced by symmetric Bayes Nash equilibrium as the admissible information structure varies, and then verify that the linked game remains in the maintained symmetric class.

3.1 The linked game

For each player i , let the linked action space be $\bar{A}_i := A^0 \times A^1$, and let $\bar{\mathcal{A}} := \prod_{i=1}^N \bar{A}_i$ denote the corresponding action-profile space. A generic linked action of player i is written $\bar{a}_i = (a_i^0, a_i^1) \in \bar{A}_i$, and a generic action profile in the linked game is written $\bar{a} = (a^0, a^1) \in \bar{\mathcal{A}}$, where $a^\ell \in \mathcal{A}^\ell, \ell \in \{0, 1\}$. The linked payoff function of player i is

$$\bar{u}_i(\bar{a}, t, \theta) := u_i^0(a^0, t, \theta) + u_i^1(a^1, t, \theta), \quad (\bar{a}, t, \theta) \in \bar{\mathcal{A}} \times T^N \times \Theta.$$

The additive payoff is an auxiliary representation device: it encodes the two component optimality requirements inside one equilibrium problem, without assuming that the payoff consequences of the two component games are physically realized together. The two action components are linked only through the common information under which $\bar{a}_i$ is chosen.

The linked action and payoff primitives are collected in $\bar{\mathcal{G}} = ((\bar{u}_i)_{i=1}^N, (\bar{A}_i)_{i=1}^N, \Theta)$. Together with the marginal distribution $\pi^{T\Theta}$ of private types and the public state, it defines the linked payoff-relevant environment

$$\bar{\mathcal{E}} := (\bar{\mathcal{G}}, \pi^{T\Theta}).$$

In the terminology of the BCE literature, $\bar{\mathcal{E}}$ is the base-game object of the linked problem: it fixes the linked action and payoff primitives and the type-state marginal $\pi^{T\Theta}$, but not the additional-signal layer.

When the linked action and payoff primitive $\bar{\mathcal{G}}$ is paired with an information structure $I = (T, (S_i)_{i=1}^N, \pi)$, the resulting linked incomplete-information game is $(\bar{\mathcal{G}}, I)$. In that game, player i observes her type t_i , signal s_i , and the public state θ , and chooses a linked action $\bar{a}_i = (a_i^0, a_i^1)$. The linked construction changes the action and payoff primitives, but

not the common information structure. The two action components are therefore chosen under the same realization of the common information structure.

3.2 Strategies and induced distributions in the linked game

Let $\mathcal{I}(\bar{\mathcal{E}})$ denote the class of admissible common information structures consistent with $\bar{\mathcal{E}}$ and satisfying the informational restrictions in Definition 1(ii)–(iii). Equivalently, these are the admissible symmetric additional-signal expansions of the linked base game $\bar{\mathcal{E}}$. Thus, for each $I = (T, (S_i)_{i=1}^N, \pi) \in \mathcal{I}(\bar{\mathcal{E}})$, the marginal of π on $T^N \times \Theta$ is $\pi^{T\Theta}$, while the additional-signal spaces $(S_i)_{i=1}^N$ and the conditional distribution of the additional-signal profile s given (t, θ) may differ across elements of $\mathcal{I}(\bar{\mathcal{E}})$.

Fix $I \in \mathcal{I}(\bar{\mathcal{E}})$. A strategy for player i in the linked game $(\bar{\mathcal{G}}, I)$ is a measurable mapping

$$\bar{\sigma}_i : T \times S_i \times \Theta \rightarrow \Delta(\bar{A}_i).$$

A strategy profile is denoted by $\bar{\sigma} = (\bar{\sigma}_1, \dots, \bar{\sigma}_N)$. Given (t, s, θ) , the profile $\bar{\sigma}$ induces the conditional distribution over action profiles

$$\bar{\sigma}(\bar{a} \mid t, s, \theta) = \prod_{i=1}^N \bar{\sigma}_i(\bar{a}_i \mid t_i, s_i, \theta), \quad \bar{a} \in \bar{\mathcal{A}}.$$

A strategy profile is symmetric if two requirements hold: first, each player's decision rule is invariant to relabeling of opponents in the opponent-indexed coordinates of s_i ; second, the same decision rule is used across players up to relabeling of player identities. A symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$ is a Bayes Nash equilibrium whose strategy profile is symmetric in this sense. Appendix A.1 records the formal definition used throughout.

For each $I \in \mathcal{I}(\bar{\mathcal{E}})$, let $\Sigma^{\text{SBNE}}(\bar{\mathcal{G}}, I)$ denote the set of symmetric Bayes Nash equilibrium strategy profiles of $(\bar{\mathcal{G}}, I)$. For $\bar{\sigma} \in \Sigma^{\text{SBNE}}(\bar{\mathcal{G}}, I)$, define the induced distribution over $\bar{\mathcal{A}} \times T^N \times \Theta$ by

$$\mu_{\bar{\sigma}, I}(\bar{a}, t, \theta) := \int_{s \in \mathcal{S}} \bar{\sigma}(\bar{a} \mid t, s, \theta) \pi(ds, t, \theta),$$

where $\mathcal{S} = \prod_{i=1}^N S_i$ is the additional-signal profile space and π is the common prior specified by I . Then

$$\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}}) := \bigcup_{I \in \mathcal{I}(\bar{\mathcal{E}})} \{ \mu_{\bar{\sigma}, I} \in \Delta(\bar{\mathcal{A}} \times T^N \times \Theta) : \bar{\sigma} \in \Sigma^{\text{SBNE}}(\bar{\mathcal{G}}, I) \}.$$

This is the set of distributions induced by symmetric Bayes Nash equilibrium in the linked game as one ranges over admissible symmetric additional-signal expansions of $\bar{\mathcal{E}}$.

The next subsection verifies that the linked game remains in the maintained symmetric class. This closure result allows the single-game SBCE characterization to be applied to the linked game, thereby characterizing $\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}})$.

3.3 Closure of the maintained class under linking

We now verify that the linked game $(\bar{\mathcal{G}}, I)$ remains in the maintained class whenever the two component games belong to that class.

Lemma 1 (Closure of payoff symmetry under linking). *If each component game $\ell \in \{0, 1\}$ satisfies Definition 1(i), then the linked game primitive $\bar{\mathcal{G}}$ also satisfies Definition 1(i).*

Proof. Fix a permutation ϕ of $\{1, \dots, N\}$, a player i , and

$$(\bar{a}, t, \theta) \in \bar{\mathcal{A}} \times T^N \times \Theta,$$

where $\bar{a} = (a^0, a^1)$, so that

$$\phi(\bar{a}) = (\phi(a^0), \phi(a^1)).$$

By definition of the linked payoff,

$$\bar{u}_{\phi(i)}(\phi(\bar{a}), \phi(t), \theta) = u_{\phi(i)}^0(\phi(a^0), \phi(t), \theta) + u_{\phi(i)}^1(\phi(a^1), \phi(t), \theta).$$

Since each component game is symmetric,

$$u_{\phi(i)}^\ell(\phi(a^\ell), \phi(t), \theta) = u_i^\ell(a^\ell, t, \theta), \quad \ell \in \{0, 1\}.$$

Hence

$$\bar{u}_{\phi(i)}(\phi(\bar{a}), \phi(t), \theta) = u_i^0(a^0, t, \theta) + u_i^1(a^1, t, \theta) = \bar{u}_i(\bar{a}, t, \theta),$$

which is the required symmetry condition. $\square$

Lemma 2 (Closure of informational restrictions under linking). *If the common information structure I satisfies Definition 1(ii)–(iii), then the linked game $(\bar{\mathcal{G}}, I)$ satisfies the same informational restrictions.*

Proof. The linked construction changes only the action and payoff primitives. It leaves the common information structure I , including the marginal $\pi^{T\Theta}$ and the conditional law of s given (t, θ) , unchanged. Thus Definition 1(ii)–(iii) continues to hold. $\square$

By Lemmas 1 and 2, the linked game $(\bar{\mathcal{G}}, I)$ belongs to the maintained symmetric latent-information class whenever the two component games belong to that class. The next section uses this closure to obtain the linked-game characterization.

4 SBCE of the linked base game and joint prediction

The previous section constructed the linked game and its base-game object

$$\bar{\mathcal{E}} = (\bar{\mathcal{G}}, \pi^{T\Theta}).$$

Throughout this section, for each admissible information structure $I \in \mathcal{I}(\bar{\mathcal{E}})$, the notation $(\bar{\mathcal{G}}, I)$ denotes the linked game generated by the expansion I , and $(\mathcal{G}^\ell, I)$ denotes the corresponding component- ℓ Bayesian game.

This section defines SBCE of the linked base game and uses the single-game SBCE characterization to obtain the linked-game equivalence. The goal is to characterize $\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}})$, the set of distributions induced by symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$ as I ranges over the admissible symmetric expansions of $\bar{\mathcal{E}}$. The resulting equivalence is then translated into a fixed-information joint-prediction statement for the two component games.

4.1 SBCE of the linked base game

We now define SBCE of the linked base game $\bar{\mathcal{E}} = (\bar{\mathcal{G}}, \pi^{T\Theta})$. This base game inherits the linked action and payoff primitives and the type-state marginal $\pi^{T\Theta}$ from the linked game, but removes the additional-signal layer from the strategic description. Thus, after (t, θ) is drawn according to $\pi^{T\Theta}$, the public state θ is observed, player i observes her private type t_i , chooses a linked action $\bar{a}_i \in \bar{A}_i$, and receives payoff $\bar{u}_i(\bar{a}, t, \theta)$.

As in BCE, one way to understand the recommendation rule is through an “omniscient” mediator. After observing the realization of (t, θ) , the mediator draws a linked action profile $\bar{a}$ and privately recommends $\bar{a}_i$ to player i . For the recommendation rule to represent equilibrium behavior in the linked base game, each player must be willing to

follow her recommended linked action conditional on her type t_i , the public state θ , and the recommendation $\bar{a}_i$. This requirement is the obedience condition defined below.

Relative to unrestricted BCE of the same base game, SBCE uses a symmetric independent recommendation rule. The mediator does not choose an arbitrary joint distribution over linked action profiles conditional on (t, θ) . Instead, recommendations are generated player by player from a common kernel. For player i , the kernel may depend on i 's own type, the public state, and an anonymous summary of opponents' types. This anonymous summary is the state variable through which opponents' types enter player i 's recommendation rule.

For each player i and each type profile $t \in T^N$, let $n_{-i}(t_{-i}) \in \mathcal{H}$ denote the vector of opponents' type counts, where

$$\mathcal{H} := \left\{ m \in \mathbb{Z}_+^{|T|} : \sum_{\tau \in T} m^\tau = N - 1 \right\}.$$

A symmetric independent recommendation rule for the linked base game is a stochastic kernel

$$\bar{\nu}(\cdot \mid t_i, m, \theta) \in \Delta(\bar{\mathcal{A}}_i), \quad (t_i, m, \theta) \in T \times \mathcal{H} \times \Theta.$$

The same kernel $\bar{\nu}$ is used for every player. Given (t, θ) , the rule induces the conditional distribution over linked action profiles

$$\bar{\nu}(\bar{a} \mid t, \theta) = \prod_{i=1}^N \bar{\nu}(\bar{a}_i \mid t_i, n_{-i}(t_{-i}), \theta), \quad \bar{a} \in \bar{\mathcal{A}}.$$

The induced distribution over $\bar{\mathcal{A}} \times T^N \times \Theta$ is

$$\mu_{\bar{\nu}}(\bar{a}, t, \theta) := \pi^{T\Theta}(t, \theta) \bar{\nu}(\bar{a} \mid t, \theta).$$

Bayes consistency. A distribution $\mu \in \Delta(\bar{\mathcal{A}} \times T^N \times \Theta)$ is *Bayes consistent* with the linked base game $\bar{\mathcal{E}}$ if its marginal on $T^N \times \Theta$ is $\pi^{T\Theta}$, that is,

$$\sum_{\bar{a} \in \bar{\mathcal{A}}} \mu(\bar{a}, t, \theta) = \pi^{T\Theta}(t, \theta), \quad \forall (t, \theta) \in T^N \times \Theta.$$

By its definition, $\mu_{\bar{\nu}}$ is Bayes consistent with $\bar{\mathcal{E}}$.

Obedience. Write $\bar{\mathcal{A}}_{-i} := \prod_{j \neq i} \bar{A}_j$ for the opponents' linked action-profile space. A distribution $\mu \in \Delta(\bar{\mathcal{A}} \times T^N \times \Theta)$ is *obedient* for the linked base game if, for every player i , every type $t_i \in T$, every recommended linked action $\bar{a}_i \in \bar{A}_i$, every public state $\theta \in \Theta$, and every alternative linked action $\bar{a}'_i \in \bar{A}_i$ such that

$$\sum_{t_{-i} \in T^{N-1}} \sum_{\bar{a}_{-i} \in \bar{\mathcal{A}}_{-i}} \mu(\bar{a}_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) > 0,$$

one has

$$\sum_{t_{-i} \in T^{N-1}} \sum_{\bar{a}_{-i} \in \bar{\mathcal{A}}_{-i}} \mu(\bar{a}_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) \left[\bar{u}_i(\bar{a}_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) - \bar{u}_i(\bar{a}'_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) \right] \geq 0.$$

Thus obedience is imposed conditional on the player's type, recommended linked action, and public state, $(t_i, \bar{a}_i, \theta)$.

A symmetric independent recommendation rule $\bar{\nu}$ is an SBCE recommendation rule for the linked base game if its induced distribution $\mu_{\bar{\nu}}$ is obedient. Bayes consistency with $\bar{\mathcal{E}}$ is satisfied by construction; obedience is the substantive restriction.

Let $\mathcal{R}^{\text{SBCE}}(\bar{\mathcal{E}})$ denote the set of SBCE recommendation rules for the linked base game $\bar{\mathcal{E}}$. For $\bar{\nu} \in \mathcal{R}^{\text{SBCE}}(\bar{\mathcal{E}})$, let $\mu_{\bar{\nu}}$ denote the distribution induced by $\bar{\nu}$. Define

$$\mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}}) := \{ \mu_{\bar{\nu}} \in \Delta(\bar{\mathcal{A}} \times T^N \times \Theta) : \bar{\nu} \in \mathcal{R}^{\text{SBCE}}(\bar{\mathcal{E}}) \}.$$

This is the set of distributions induced by SBCE of the linked base game.

4.2 Exact characterization of the linked problem

This subsection supplies the linked-game equivalence used below. Its role is to bring the single-game SBCE characterization to the linked problem through the linked base game; the fixed-information content enters in the joint-prediction and counterfactual projection results that follow.

The outcome set to be characterized is $\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}})$: the set of distributions induced by symmetric Bayes Nash equilibrium over admissible symmetric expansions of the linked base game. Even within the maintained symmetric class, this object is not convenient to characterize directly, because it requires ranging over a latent, and possibly infinite-dimensional, additional-signal layer. The role of SBCE is to replace this search over expansions by an equilibrium object defined directly on the linked base game.

This is the same base-game logic that underlies the BCE literature. For a given base game, equilibrium behavior over expansions is characterized by a correlated-equilibrium object defined on that base game. In the symmetric latent-information class studied by Lu (2026), the corresponding object is SBCE: for each given payoff-relevant environment in the class, SBCE characterizes the distributions induced by symmetric Bayes Nash equilibrium over admissible symmetric additional-signal expansions. This subsection brings that base-game characterization logic to the linked problem through the linked base game $\bar{\mathcal{E}}$.

The remaining step is to verify that the linked problem falls within the scope of the single-game characterization. This is exactly the role of Section 3.3. Lemmas 1 and 2 show that, whenever the two component games belong to the maintained symmetric latent-information class, the linked game $(\bar{\mathcal{G}}, I)$ also belongs to that class. Hence the linked problem constructed above falls within the domain of the single-game SBCE characterization.

Proposition 1 (Exact characterization of the linked problem). *Suppose that each component game $\ell \in \{0, 1\}$ belongs to the maintained symmetric latent-information class. Then*

$$\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}}) = \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}}).$$

Proof. By Lemmas 1 and 2, for every admissible information structure $I \in \mathcal{I}(\bar{\mathcal{E}})$, the expanded linked game $(\bar{\mathcal{G}}, I)$ satisfies the maintained symmetric latent-information restrictions whenever the two component games do. The linked construction has common finite single-player action space $D := A^0 \times A^1$, so that $\bar{A}_i = D$ for every player i , and the associated linked base game is $\bar{\mathcal{E}} = (\bar{\mathcal{G}}, \pi^{T\Theta})$.

Thus the linked problem satisfies the hypotheses of the single-game exact characterization of Lu (2026, Theorem 1). That theorem states that, for a fixed base-game object, SBCE characterizes the distributions induced by symmetric Bayes Nash equilibrium as the admissible symmetric latent-information expansion varies. Applying that characterization to the linked problem through $\bar{\mathcal{E}}$, and using the definitions of $\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}})$ and $\mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}})$, gives

$$\mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}}) = \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}}).$$

□

Proposition 1 provides the reduced-form description of the distributions generated by admissible symmetric expansions of the linked problem. The next subsection uses this equivalence to characterize pairs of component distributions that can be generated under one common information structure.

4.3 Joint prediction under fixed information

The equivalence in Proposition 1 is a statement about outcome distributions in the linked game. To obtain a fixed-information joint-prediction statement, we relate equilibria of the linked game to equilibria of the two component games under a common information structure. A *joint prediction* under fixed information is a pair of component distributions generated by symmetric Bayes Nash equilibria under the same admissible information structure.

Definition 2 (Joint prediction under fixed information). *A pair*

$$(\mu^0, \mu^1) \in \Delta(\mathcal{A}^0 \times T^N \times \Theta) \times \Delta(\mathcal{A}^1 \times T^N \times \Theta)$$

is a joint prediction for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information if there exists $I \in \mathcal{I}(\bar{\mathcal{E}})$ and, for each $\ell \in \{0, 1\}$, a symmetric Bayes Nash equilibrium σ^ℓ of $(\mathcal{G}^\ell, I)$ such that the distribution induced by σ^ℓ in $(\mathcal{G}^\ell, I)$ is μ^ℓ .

The next lemma gives the fixed-information interpretation of the linked game. It relates equilibria of the linked game $(\bar{\mathcal{G}}, I)$ to equilibria of the two component games $(\mathcal{G}^0, I)$ and $(\mathcal{G}^1, I)$ under the same information structure.

Lemma 3 (Linked-equilibrium bridge). *Fix $I \in \mathcal{I}(\bar{\mathcal{E}})$.*

1. *If, for each $\ell \in \{0, 1\}$, σ^ℓ is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$, then the strategy profile $\bar{\sigma}$ in the linked game defined by*

$$\bar{\sigma}_i(\bar{a}_i \mid t_i, s_i, \theta) := \sigma_i^0(a_i^0 \mid t_i, s_i, \theta) \sigma_i^1(a_i^1 \mid t_i, s_i, \theta), \quad \bar{a}_i = (a_i^0, a_i^1) \in \bar{A}_i,$$

is a symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$.

2. *Conversely, if $\bar{\sigma}$ is a symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$, then for each $\ell \in \{0, 1\}$, the component marginal strategy profile σ^ℓ defined by*

$$\sigma_i^\ell(a_i^\ell \mid t_i, s_i, \theta) := \sum_{a_i^{1-\ell} \in A^{1-\ell}} \bar{\sigma}_i(a_i^0, a_i^1 \mid t_i, s_i, \theta)$$

is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$.

Proof. The key point is that linked payoffs are additively separable across the two components. Hence the expected payoff from any linked strategy equals the sum of the expected component payoffs generated by its two component marginals.

For the first claim, suppose that σ^ℓ is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$ for each $\ell \in \{0, 1\}$. Consider any linked deviation by player i . Its two component marginals are feasible deviations in the two component games. Since each σ^ℓ is optimal in $(\mathcal{G}^\ell, I)$, neither component marginal can increase the corresponding component payoff. Therefore the linked deviation cannot increase the sum of component payoffs. The product construction also preserves symmetry, so $\bar{\sigma}$ is a symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$.

For the converse, suppose that $\bar{\sigma}$ is a symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$. If, for some ℓ , the component marginal σ^ℓ were not optimal in $(\mathcal{G}^\ell, I)$, then player i could profitably deviate in the linked game by changing the ℓ -component of her linked strategy while keeping the other component marginal unchanged. This would increase the linked payoff, contradicting optimality of $\bar{\sigma}$. Symmetry of the component marginals follows from symmetry of $\bar{\sigma}$. Hence each σ^ℓ is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$.

Appendix A.2 gives the formal proof. $\square$

Thus, at the component-marginal level relevant for joint prediction, the linked equilibrium problem is equivalent to the original fixed-information problem. The next theorem combines this bridge with Proposition 1.

Theorem 1 (SBCE joint prediction under fixed information). *Suppose that each component game $\ell \in \{0, 1\}$ belongs to the maintained symmetric latent-information class. Then a pair*

$$(\mu^0, \mu^1) \in \Delta(\mathcal{A}^0 \times T^N \times \Theta) \times \Delta(\mathcal{A}^1 \times T^N \times \Theta)$$

is a joint prediction for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information if and only if there exists $\bar{\mu} \in \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}})$ such that $\text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\bar{\mu}) = \mu^0$ and $\text{proj}_{\mathcal{A}^1 \times T^N \times \Theta}(\bar{\mu}) = \mu^1$.

Proof. Suppose first that (μ^0, μ^1) is a joint prediction under fixed information. Then there exists a common information structure $I = (T, (S_i)_{i=1}^N, \pi)$ admissible for the maintained class and, for each $\ell \in \{0, 1\}$, a symmetric Bayes Nash equilibrium σ^ℓ of $(\mathcal{G}^\ell, I)$ such that the distribution induced by σ^ℓ in $(\mathcal{G}^\ell, I)$ is μ^ℓ .

For each player i , define a linked strategy by

$$\bar{\sigma}_i(\bar{a}_i \mid t_i, s_i, \theta) = \sigma_i^0(a_i^0 \mid t_i, s_i, \theta) \sigma_i^1(a_i^1 \mid t_i, s_i, \theta), \quad \bar{a}_i = (a_i^0, a_i^1) \in \bar{A}_i.$$

The resulting profile $\bar{\sigma} = (\bar{\sigma}_1, \dots, \bar{\sigma}_N)$ is symmetric and, by Lemma 3, is a symmetric Bayes Nash equilibrium of the linked game $(\bar{\mathcal{G}}, I)$. Let $\bar{\mu} := \mu_{\bar{\sigma}, I}$ denote the induced distribution. By construction, $\text{proj}_{\mathcal{A}^\ell \times T^N \times \Theta}(\bar{\mu}) = \mu^\ell, \ell \in \{0, 1\}$. Since $\bar{\mu} \in \mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}})$, Proposition 1 implies $\bar{\mu} \in \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}})$.

Conversely, suppose there exists $\bar{\mu} \in \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}})$ such that $\text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\bar{\mu}) = \mu^0$ and $\text{proj}_{\mathcal{A}^1 \times T^N \times \Theta}(\bar{\mu}) = \mu^1$. By Proposition 1, $\bar{\mu} \in \mathcal{M}^{\text{SBNE}}(\bar{\mathcal{E}})$. Hence there exists a common information structure $I = (T, (S_i)_{i=1}^N, \pi)$ admissible for the maintained class and a symmetric Bayes Nash equilibrium $\bar{\sigma}$ of the linked game $(\bar{\mathcal{G}}, I)$ such that $\bar{\mu} = \mu_{\bar{\sigma}, I}$.

For each $\ell \in \{0, 1\}$, let σ^ℓ denote the component- ℓ marginal strategy profile induced by $\bar{\sigma}$. By Lemma 3, each σ^ℓ is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$. By construction, the distribution induced by σ^ℓ in $(\mathcal{G}^\ell, I)$ is exactly $\text{proj}_{\mathcal{A}^\ell \times T^N \times \Theta}(\bar{\mu}) = \mu^\ell$. Therefore (μ^0, μ^1) is a joint prediction under fixed information. $\square$

Theorem 1 shows that joint predictions for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information are characterized exactly by the component projections of distributions induced by SBCE of the linked base game.

5 Fixed-information counterfactuals

We now specialize the linked analysis to counterfactual prediction. Game $\ell = 0$ is treated as the observed game, and game $\ell = 1$ as the counterfactual game. Under the fixed-information restriction, the observed and counterfactual distributions must be jointly rationalized under one admissible information structure. The counterfactual characterization below implements this requirement by matching the observed component of a distribution induced by SBCE of the linked base game and then projecting onto the counterfactual component.

5.1 The fixed-information counterfactual set

Fix an observed component-game distribution $\mu^{\text{obs}} \in \Delta(\mathcal{A}^0 \times T^N \times \Theta)$.

Definition 3 (Counterfactual prediction under fixed information). *A distribution $\mu^{\text{cf}} \in \Delta(\mathcal{A}^1 \times T^N \times \Theta)$ is a counterfactual prediction for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information given μ^{obs} if there exists $I \in \mathcal{I}(\bar{\mathcal{E}})$ and, for each $\ell \in \{0, 1\}$, a symmetric Bayes Nash equilibrium σ^ℓ of $(\mathcal{G}^\ell, I)$ such that the distribution induced by σ^0 in $(\mathcal{G}^0, I)$ is μ^{obs} and the distribution induced by σ^1 in $(\mathcal{G}^1, I)$ is μ^{cf} .*

Thus the observed and counterfactual distributions must be jointly rationalized under a common admissible information structure in the maintained class. Let $\mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}})$ denote the set of counterfactual predictions for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information given μ^{obs} .

By Theorem 1, this joint-rationalization requirement has an equivalent representation through SBCE of the linked base game. A distribution $\bar{\mu} \in \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}})$ is feasible for μ^{obs} if $\text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\bar{\mu}) = \mu^{\text{obs}}$. Accordingly, define the feasible set

$$\mathcal{F}(\mu^{\text{obs}}) := \left\{ \bar{\mu} \in \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}}) : \text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\bar{\mu}) = \mu^{\text{obs}} \right\}.$$

5.2 Counterfactual projection

Theorem 2 (SBCE counterfactual projection). *Suppose that each component game $\ell \in \{0, 1\}$ belongs to the maintained symmetric latent-information class. Fix*

$$\mu^{\text{obs}} \in \Delta(\mathcal{A}^0 \times T^N \times \Theta).$$

Then

$$\mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}}) = \text{proj}_{\mathcal{A}^1 \times T^N \times \Theta} \mathcal{F}(\mu^{\text{obs}}).$$

Equivalently, $\mu^{\text{cf}} \in \mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}})$ if and only if there exists $\bar{\mu} \in \mathcal{F}(\mu^{\text{obs}})$ such that

$$\text{proj}_{\mathcal{A}^1 \times T^N \times \Theta}(\bar{\mu}) = \mu^{\text{cf}}.$$

Proof. By Definition 3, μ^{cf} is a counterfactual prediction for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information given μ^{obs} if and only if $(\mu^{\text{obs}}, \mu^{\text{cf}})$ is a joint prediction for $(\mathcal{E}^0, \mathcal{E}^1)$ under fixed information. By Theorem 1, this holds if and only if there exists $\bar{\mu} \in \mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}})$ such that

$$\text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\bar{\mu}) = \mu^{\text{obs}} \quad \text{and} \quad \text{proj}_{\mathcal{A}^1 \times T^N \times \Theta}(\bar{\mu}) = \mu^{\text{cf}}.$$

The first equality is equivalent to $\bar{\mu} \in \mathcal{F}(\mu^{\text{obs}})$, and the second is exactly the requirement that μ^{cf} be the counterfactual projection of some distribution in $\mathcal{F}(\mu^{\text{obs}})$. This proves the characterization. $\square$

Thus the fixed-information counterfactual set is exactly the projection, onto the counterfactual component, of the distributions induced by SBCE of the linked base game that match the observed marginal. Theorem 2 states the characterization at the level of distributions over $\bar{\mathcal{A}} \times T^N \times \Theta$. Appendix B gives the corresponding primitive representation of the feasible set $\mathcal{F}(\mu^{\text{obs}})$. In that representation, one searches over a symmetric independent recommendation rule for the linked base game,

$$\bar{\nu}(\bar{a}_i \mid t_i, n_{-i}(t_{-i}), \theta),$$

and the distribution over $\bar{\mathcal{A}} \times T^N \times \Theta$ is generated from this lower-dimensional primitive object through the product structure of the recommendation rule. This makes explicit the primitive restrictions that implement the projection characterization.

6 Comparison with unrestricted BCE counterfactuals

The fixed-information counterfactual characterization above is derived within the maintained symmetric latent-information class. It is therefore natural to compare the resulting counterfactual set with the unrestricted BCE counterfactual set in the linked-game construction of Bergemann et al. (2022), applied to the same linked base game. This comparison is between two maintained ways of carrying the same linked base game into fixed-information counterfactual analysis, not between an unrestricted object and a refinement of that object. Both constructions fix the same linked base game $\bar{\mathcal{E}}$ and hence inherit the payoff and type-state restrictions already encoded in that base game, including payoff symmetry and exchangeability of private types. The difference is that SBCE of the linked base game also reflects the maintained restrictions that are not encoded by the linked base game itself: symmetric equilibrium behavior and symmetric conditional independence in the latent additional-signal layer.

The distinction is at the level of how distributions over $\bar{\mathcal{A}} \times T^N \times \Theta$ may be generated. Under unrestricted BCE, such a distribution must satisfy Bayes consistency with the linked base game and obedience for the linked game. It need not be generated by an admissible symmetric additional-signal expansion. Under SBCE, a feasible distribution must be induced by a symmetric independent recommendation rule for the linked base game. Thus the joint distribution over linked actions, private types, and the public state must

arise from a common player-wise recommendation kernel and the associated product structure. Appendix B gives the primitive representation of the SBCE feasible set, including the observed-marginal matching and obedience constraints.

This difference matters because unrestricted BCE can encode cross-player correlations in linked action recommendations directly at the level of the joint distribution over linked actions, private types, and the public state. SBCE of the linked base game does not allow an arbitrary joint recommendation rule. Conditional on the type-state profile, recommendations must be generated player by player from a common symmetric kernel. Thus some cross-player correlations in linked action recommendations that are feasible under unrestricted BCE need not be implementable by admissible symmetric additional-signal expansions. The entry environment below makes this mechanism explicit.

6.1 Unrestricted BCE feasible sets

Let $\mathcal{M}^{\text{BCE}}(\bar{\mathcal{E}})$ denote the set of unrestricted BCE distributions of the linked base game $\bar{\mathcal{E}} = (\bar{\mathcal{G}}, \pi^{T\Theta})$. This set imposes Bayes consistency with $\pi^{T\Theta}$ and obedience for the linked game, but does not require the distribution to be induced by a symmetric independent recommendation rule.

For a given observed component distribution $\mu^{\text{obs}} \in \Delta(\mathcal{A}^0 \times T^N \times \Theta)$, define the unrestricted BCE feasible set

$$\mathcal{F}^{\text{BCE}}(\mu^{\text{obs}}) := \left\{ \bar{\mu} \in \mathcal{M}^{\text{BCE}}(\bar{\mathcal{E}}) : \text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\bar{\mu}) = \mu^{\text{obs}} \right\},$$

and the corresponding unrestricted BCE counterfactual set

$$\mathcal{C}^{\text{BCE}}(\mu^{\text{obs}}) := \text{proj}_{\mathcal{A}^1 \times T^N \times \Theta} \mathcal{F}^{\text{BCE}}(\mu^{\text{obs}}).$$

6.2 Nestedness relative to unrestricted BCE

Proposition 2 (Nestedness relative to unrestricted BCE). *For every observed component distribution $\mu^{\text{obs}} \in \Delta(\mathcal{A}^0 \times T^N \times \Theta)$, the SBCE counterfactual set is contained in the unrestricted BCE counterfactual set:*

$$\mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}}) \subseteq \mathcal{C}^{\text{BCE}}(\mu^{\text{obs}}).$$

Proof. Every distribution induced by SBCE of the linked base game is Bayes consistent with $\pi^{T\Theta}$ and satisfies obedience for the linked game. Hence $\mathcal{M}^{\text{SBCE}}(\bar{\mathcal{E}}) \subseteq \mathcal{M}^{\text{BCE}}(\bar{\mathcal{E}})$.

Imposing the same observed-marginal restriction on both sides gives $\mathcal{F}(\mu^{\text{obs}}) \subseteq \mathcal{F}^{\text{BCE}}(\mu^{\text{obs}})$. Projecting this inclusion onto the counterfactual component yields

$$\mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}}) = \text{proj}_{\mathcal{A}^1 \times T^N \times \Theta} \mathcal{F}(\mu^{\text{obs}}) \subseteq \text{proj}_{\mathcal{A}^1 \times T^N \times \Theta} \mathcal{F}^{\text{BCE}}(\mu^{\text{obs}}) = \mathcal{C}^{\text{BCE}}(\mu^{\text{obs}}).$$

□

Proposition 2 shows that the primitive restrictions of the maintained symmetric class weakly tighten the fixed-information counterfactual set relative to unrestricted BCE of the same linked base game. The next question is whether this tightening can matter for an economic statistic of interest, rather than only for the set of feasible distributions. The two-firm entry environment below shows that it can: the tightening changes the producer-surplus comparative static.

7 The two-firm entry environment

This section revisits the two-firm entry environment of Bergemann et al. (2022) to show that the nestedness in Proposition 2 can be strict for an economically relevant statistic. For each counterfactual parameter z , the SBCE and unrestricted BCE counterfactual sets are constructed from the same linked base game $\bar{\mathcal{E}}_z$ and the same observed marginal. The distinction is again that SBCE of the linked base game also reflects the maintained restrictions not already encoded by the linked base game itself: symmetric equilibrium behavior and symmetric conditional independence in the latent additional-signal layer.

Environment and observed outcome. There are two firms, indexed by $i \in \{1, 2\}$. In each component game, $A^0 = A^1 = \{E, N\}$, where E denotes entry and N denotes nonentry. The private cost-type space is $T = \{0, 2\}$, where $c_i \in T$ is firm i 's private cost type. There is no nontrivial public state, so we suppress θ . The cost-type profile $c = (c_1, c_2) \in T^2$ is uniformly distributed on $\{0, 2\}^2$.

In the observed game, monopoly and duopoly entry payoffs are $3 - c_i$ and $1 - c_i$, respectively, while nonentry yields zero. The observed distribution is the one in which each firm enters if and only if its own cost is low:

$$\mu^{\text{obs}}((E, E), (0, 0)) = \mu^{\text{obs}}((E, N), (0, 2)) = \mu^{\text{obs}}((N, E), (2, 0)) = \mu^{\text{obs}}((N, N), (2, 2)) = \frac{1}{4},$$

with zero mass on all other observed action–cost profiles.

As in Bergemann et al. (2022), this observed distribution is induced by the symmetric equilibrium under the information structure in which each firm observes only its own cost and receives no additional private signal. Consider the symmetric pure strategy

$$\sigma_i(E \mid c_i = 0) = 1, \quad \sigma_i(N \mid c_i = 2) = 1, \quad i = 1, 2.$$

Under this strategy, a low-cost firm strictly prefers entry: conditional on $c_i = 0$, the opponent is low cost with probability $1/2$ and high cost with probability $1/2$, so the expected payoff from entry is

$$\frac{1}{2}(1) + \frac{1}{2}(3) = 2 > 0.$$

A high-cost firm is indifferent between entry and nonentry: conditional on $c_i = 2$, the expected payoff from entry is

$$\frac{1}{2}(-1) + \frac{1}{2}(1) = 0.$$

Thus σ is a symmetric Bayes Nash equilibrium and induces μ^{obs} . The analyst takes μ^{obs} as the observed component but does not observe the information structure that generated it.

The fixed-information counterfactual constructions below range over information structures that rationalize this observed distribution and are held common across the observed and counterfactual components. Under SBCE, equilibrium behavior is restricted to be symmetric, and the additional-signal layer is restricted to the maintained symmetric class: conditional on the cost profile, additional private signals are generated from a common symmetric kernel and are conditionally independent across firms. Under unrestricted BCE, firms may have access to additional information beyond their private cost types, but no symmetric conditional-independence restriction is imposed on the additional-signal layer.

Counterfactual sets. In the counterfactual game indexed by z , monopoly and duopoly entry payoffs are $3 + z - c_i$ and $1 + z - c_i$. For each z , let $\bar{\mathcal{E}}_z$ denote the linked base game obtained by linking the observed game and the counterfactual game indexed by z . Write $\mathcal{C}_z^{\text{SBCE}}(\mu^{\text{obs}})$ and $\mathcal{C}_z^{\text{BCE}}(\mu^{\text{obs}})$ for the corresponding SBCE and unrestricted BCE counterfactual sets. For any counterfactual distribution μ^{cf} , let $W(\mu^{\text{cf}})$ denote the associated expected producer surplus. Define the SBCE producer-surplus prediction set by

$$\mathcal{W}^{\text{SBCE}}(z) := \{W(\mu^{\text{cf}}) : \mu^{\text{cf}} \in \mathcal{C}_z^{\text{SBCE}}(\mu^{\text{obs}})\}.$$

Define $\mathcal{W}^{\text{BCE}}(z)$ analogously using $\mathcal{C}_z^{\text{BCE}}(\mu^{\text{obs}})$. For notational simplicity, the dependence on the given observed distribution μ^{obs} is suppressed in $\mathcal{W}^{\text{SBCE}}(z)$ and $\mathcal{W}^{\text{BCE}}(z)$.

The economic question is simple. The counterfactual parameter z raises the payoff from entry in both monopoly and duopoly states. Does this payoff shifter raise producer surplus? Under unrestricted BCE of the same linked base game, the answer is set-valued: for $z \in (0, 1)$, producer surplus can be above or below the level generated by the symmetric mixed-strategy analysis in Bergemann et al. (2022).

Under SBCE of the linked base game, by contrast, the answer is a singleton. For $z \in [0, 1]$, the producer-surplus prediction set is

$$\mathcal{W}^{\text{SBCE}}(z) = \{2\}.$$

The neutrality is not because the payoff shifter is irrelevant. The higher payoff from entry induces additional high-cost entry, and the resulting adjustment in market structure exactly offsets the direct payoff increase. The rest of the section derives this prediction and compares it with the unrestricted BCE prediction set.

Because there are two firms, the anonymous summary of the opponent's type is simply the opponent's cost type. After imposing the observed marginal, the observed component of the linked recommendation is pinned down almost surely: a low-cost firm is recommended to enter in the observed game, and a high-cost firm is recommended not to enter. Hence an SBCE recommendation rule for the linked base game can be summarized by four counterfactual entry probabilities:

$$\begin{aligned} p_{00} &:= \Pr(a_i^1 = E \mid c_i = 0, c_{-i} = 0), & p_{02} &:= \Pr(a_i^1 = E \mid c_i = 0, c_{-i} = 2), \\ p_{20} &:= \Pr(a_i^1 = E \mid c_i = 2, c_{-i} = 0), & p_{22} &:= \Pr(a_i^1 = E \mid c_i = 2, c_{-i} = 2). \end{aligned}$$

Proposition 3 (Producer-surplus neutrality in the entry environment). *For every $z \in [0, 1]$, the SBCE counterfactual set in the two-firm entry environment with binary cost types is a singleton. The unique feasible SBCE recommendation rule satisfies*

$$p_{00} = 1, \quad p_{02} = 1, \quad p_{20} = z, \quad p_{22} = z.$$

Consequently, the SBCE producer-surplus prediction set is

$$\mathcal{W}^{\text{SBCE}}(z) = \{2\}.$$

Proof. See Appendix A.3.

The intuition is as follows. Low-cost firms strictly prefer entry in the counterfactual component for every $z \in [0, 1]$, so they must enter. For high-cost firms, the observed marginal pins down the observed recommendation. Obedience for the linked base game, together with the product structure of the symmetric independent recommendation rule, then rules out a counterfactual recommendation that changes posterior beliefs about the opponent's cost. Hence the high-cost counterfactual entry probability must be the same whether the opponent is low cost or high cost: $p_{20} = p_{22}$. Let $p_H := p_{20} = p_{22}$. Given this equality, a high-cost firm's counterfactual recommendation does not change its posterior over the opponent's cost: the opponent is low cost with probability $1/2$ and high cost with probability $1/2$. If the opponent is low cost, it enters for sure, so the high-cost firm's counterfactual entry payoff is $z - 1$. If the opponent is high cost, it enters with probability p_H , so the high-cost firm's expected counterfactual entry payoff is

$$(1 - p_H)(1 + z) + p_H(z - 1) = 1 + z - 2p_H.$$

Hence the high-cost firm's expected gain from counterfactual entry is

$$\frac{1}{2}(z - 1) + \frac{1}{2}(1 + z - 2p_H) = z - p_H.$$

If $p_H > z$, entry recommendations occur with positive probability and are not obedient. If $p_H < z$, nonentry recommendations occur with positive probability and are not obedient. Therefore $p_H = z$ for every $z \in [0, 1]$. Thus z does affect behavior: it increases the counterfactual entry probability of high-cost firms. In this entry environment, however, the induced adjustment in market structure exactly offsets the direct increase in entry payoffs, leaving total producer surplus equal to 2.

Figure 1 plots the resulting producer-surplus predictions. The horizontal axis is extended beyond $z \in [0, 1]$ to show that the strictness result is concentrated on the intermediate range $z \in [0, 1]$.

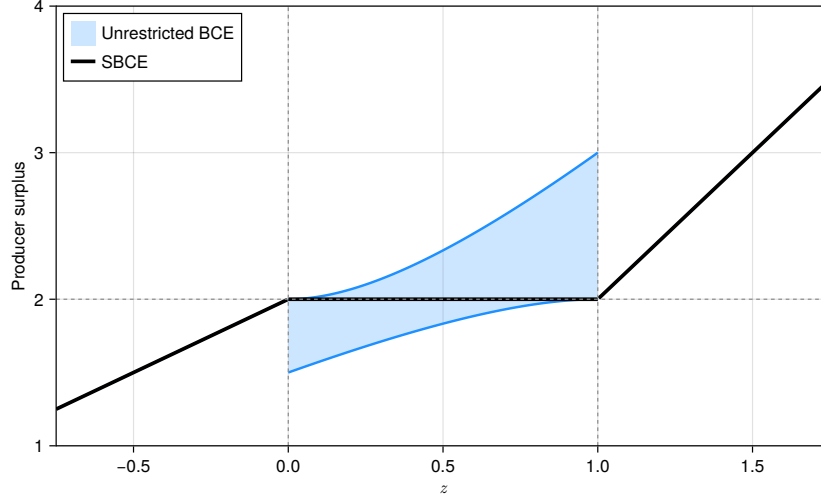


Figure 1. PRODUCER-SURPLUS PREDICTIONS IN THE TWO-FIRM ENTRY ENVIRONMENT. The shaded region is the unrestricted BCE prediction set for the linked base game, and the solid curve is the SBCE prediction. For $z \in [0, 1]$, SBCE yields $\mathcal{W}^{\text{SBCE}}(z) = \{2\}$. On the interior $z \in (0, 1)$, unrestricted BCE permits producer-surplus predictions both above and below 2.

The neutrality prediction coincides with the symmetric mixed-strategy producer surplus in Bergemann et al. (2022). The reason is not that SBCE selects this outcome from the unrestricted BCE set. Rather, once the maintained symmetric conditional-independence restriction on the additional-signal layer is imposed, the correlated high-cost recommendation patterns that generate the unrestricted producer-surplus extremes are no longer implementable by admissible symmetric expansions. Proposition 3 therefore reflects a primitive restriction on the admissible latent additional-signal structures, not an equilibrium-selection criterion.

The comparison with unrestricted BCE makes clear why the restrictions implied by the maintained class matter. In the unrestricted fixed-information analysis of Bergemann et al. (2022), the producer-surplus prediction set for $z \in [0, 1]$ is

$$\mathcal{W}^{\text{BCE}}(z) = \left[2 + z - \frac{1}{2-z}, 2 + \frac{2z^2}{1+z} \right],$$

while the symmetric mixed-strategy analysis gives producer surplus 2. By Proposition 3, $\mathcal{W}^{\text{SBCE}}(z) = \{2\}$. For every $z \in (0, 1)$,

$$2 + z - \frac{1}{2-z} < 2 < 2 + \frac{2z^2}{1+z}.$$

Thus, on the interior of the parameter range, the SBCE prediction $\{2\}$ lies strictly inside the unrestricted producer-surplus prediction set. At the endpoints, the SBCE prediction lies on the boundary of the unrestricted set, but the containment remains strict: the unrestricted set is $[3/2, 2]$ at $z = 0$ and $[2, 3]$ at $z = 1$. Hence the inclusion in Proposition 2 is strict for every $z \in [0, 1]$ in this two-firm entry environment.

Extended domain. The expanded domain in Figure 1 is included only to locate the strictness result. For $z \in (-1, 0)$, both unrestricted BCE and SBCE imply that firms enter if and only if their costs are low, so

$$\mathcal{W}^{\text{BCE}}(z) = \mathcal{W}^{\text{SBCE}}(z) = \{2 + z\}.$$

For $z > 1$, entry is strictly dominant for both cost types, so both prediction sets imply that both firms enter in every cost state, and hence

$$\mathcal{W}^{\text{BCE}}(z) = \mathcal{W}^{\text{SBCE}}(z) = \{2z\}.$$

Thus the two producer-surplus prediction sets coincide on the open side ranges. The distinction in this example is concentrated in the intermediate range $z \in [0, 1]$.

The entry environment shows that the maintained restrictions on the additional-signal layer can have welfare content. Under unrestricted BCE, the payoff shifter z is compatible with producer surplus moving above or below 2. Under SBCE of the linked base game, the same payoff shifter yields the singleton prediction: $\mathcal{W}^{\text{SBCE}}(z) = \{2\}$.

8 Conclusion

This paper studies fixed-information counterfactual analysis when the analyst maintains primitive restrictions on the latent information environment. Starting from an observed game and a counterfactual game that share one latent information structure, the paper constructs a linked base game and characterizes the distributions generated by admissible symmetric expansions of the linked problem. The central point is that the restrictions defining the class survive linking and therefore constrain counterfactual predictions.

This characterization turns fixed-information counterfactual prediction into a projection problem. Once the observed marginal is given, the admissible counterfactual distributions

are the counterfactual projections of distributions induced by SBCE of the linked base game that match the observed component. Thus the linked construction preserves the logic of fixed-information counterfactual analysis while carrying into the counterfactual problem the restrictions implied by the maintained symmetric class.

The comparison with unrestricted BCE clarifies the role of these restrictions. Both counterfactual constructions fix the same linked base game and impose the same observed-marginal restriction. Hence both inherit the payoff and type-state restrictions encoded in the linked base game, including payoff symmetry and exchangeability of private types. The difference is that SBCE of the linked base game also reflects the maintained restrictions not encoded by the linked base game itself: symmetric equilibrium behavior and symmetric conditional independence in the latent additional-signal layer. As a result, the SBCE counterfactual set is contained in the unrestricted BCE counterfactual set.

The two-firm entry environment illustrates that this containment can have economic content. In that environment, unrestricted BCE permits producer-surplus predictions both above and below 2 for $z \in (0, 1)$, whereas SBCE yields the singleton producer-surplus prediction $\mathcal{W}^{\text{SBCE}}(z) = \{2\}$. The neutrality result is not driven by the irrelevance of the payoff shifter. Rather, the payoff increase induces additional high-cost entry, and the resulting adjustment in market structure exactly offsets the direct payoff effect.

The contribution is not to propose SBCE as a general-purpose substitute for unrestricted BCE. Unrestricted BCE remains the natural object when the analyst fixes the linked base game but does not maintain the symmetric latent-information class studied here. The broader lesson is that fixed information alone is not the only source of counterfactual discipline. When the analyst maintains primitive restrictions on the latent information environment, those restrictions have independent counterfactual content and can change welfare conclusions.

A Formal equilibrium definitions and proofs

A.1 Symmetric Bayes Nash equilibrium

This subsection records the formal equilibrium definition used for both component games and the linked game. Consider a generic game in the maintained class with common finite action set D , action-profile space

$$\mathcal{D} := D^N,$$

payoff functions

$$\tilde{u}_i : \mathcal{D} \times T^N \times \Theta \rightarrow \mathbb{R},$$

and information structure

$$I = (T, (S_i)_{i=1}^N, \pi).$$

For a component game, $D = A^\ell$ and $\mathcal{D} = \mathcal{A}^\ell$. For the linked game, $D = \bar{A}_i = A^0 \times A^1$ for every player i , and $\mathcal{D} = \bar{\mathcal{A}}$.

In such a game, player i observes her private information and the public state,

$$(t_i, s_i, \theta) \in T \times S_i \times \Theta,$$

before choosing an action. A mixed strategy for player i is a measurable mapping

$$\sigma_i : T \times S_i \times \Theta \rightarrow \Delta(D).$$

A strategy profile is denoted by

$$\sigma = (\sigma_1, \dots, \sigma_N).$$

Given $(t, s, \theta) \in T^N \times \mathcal{S} \times \Theta$, the profile σ induces the product distribution

$$\sigma(d \mid t, s, \theta) := \prod_{i=1}^N \sigma_i(d_i \mid t_i, s_i, \theta), \quad d = (d_1, \dots, d_N) \in \mathcal{D}.$$

We next formalize symmetry of the strategy profile. For each player i , let φ_{-i} denote a permutation of player i 's opponents, acting on the opponent-indexed coordinates of s_i . A strategy profile σ is *symmetric* if it satisfies the following two conditions.

- (i) *Invariance to opponent relabeling within each player.* For every player i , every type $t_i \in T$, every additional signal $s_i \in S_i$, every public state $\theta \in \Theta$, every action $d_i \in D$, and every permutation φ_{-i} of player i 's opponents,

$$\sigma_i(d_i \mid t_i, s_i, \theta) = \sigma_i(d_i \mid t_i, \varphi_{-i}(s_i), \theta).$$

- (ii) *Common decision rule across players.* For any two players i and j , let $\psi_{ij} : S_i \rightarrow S_j$ denote the natural relabeling of opponent-indexed coordinates induced by the relabeling of player identities from i to j . Then, for every type $t_i \in T$, every additional signal $s_i \in S_i$, every public state $\theta \in \Theta$, and every action $d_i \in D$,

$$\sigma_i(d_i \mid t_i, s_i, \theta) = \sigma_j(d_i \mid t_i, \psi_{ij}(s_i), \theta).$$

Equivalently, all players use the same conditional law of action, up to relabeling of player identities and corresponding opponent-indexed coordinates.

Player i 's expected payoff under σ is

$$U_i(\sigma) = \sum_{\theta \in \Theta} \sum_{t \in T^N} \int_{s \in \mathcal{S}} \sum_{d \in \mathcal{D}} \tilde{u}_i(d, t, \theta) \sigma(d \mid t, s, \theta) \pi(ds, t, \theta).$$

A symmetric strategy profile σ is a *symmetric Bayes Nash equilibrium* if, for every player i and every measurable deviation

$$\tilde{\sigma}_i : T \times S_i \times \Theta \rightarrow \Delta(D),$$

one has

$$U_i(\sigma) \geq U_i(\tilde{\sigma}_i, \sigma_{-i}),$$

where $U_i(\tilde{\sigma}_i, \sigma_{-i})$ is defined by replacing σ_i with $\tilde{\sigma}_i$ in the product distribution above.

A.2 Proof of Lemma 3

Proof. For each player i , let $\bar{U}_i(\bar{\sigma})$ denote expected payoff in the linked game $(\bar{\mathcal{G}}, I)$, and let

$$U_i^\ell(\sigma^\ell), \quad \ell \in \{0, 1\},$$

denote expected payoff in the component game $(\mathcal{G}^\ell, I)$. Because the linked payoff is additively separable,

$$\bar{u}_i(\bar{a}, t, \theta) = u_i^0(a^0, t, \theta) + u_i^1(a^1, t, \theta),$$

expected payoff in the linked game decomposes as

$$\bar{U}_i(\bar{\sigma}) = U_i^0(\sigma^0) + U_i^1(\sigma^1),$$

whenever σ^ℓ is the component- ℓ marginal strategy profile induced by $\bar{\sigma}$.

We first prove part (1). Suppose that σ^ℓ is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$ for each $\ell \in \{0, 1\}$, and define $\bar{\sigma}$ by the product formula in the statement.

Symmetry of $\bar{\sigma}$ follows from symmetry of each σ^ℓ . To verify optimality, fix a player i and an alternative linked strategy $\bar{\tau}_i$. Let τ_i^ℓ denote its component marginal,

$$\tau_i^\ell(a_i^\ell \mid t_i, s_i, \theta) := \sum_{a_i^{1-\ell} \in A^{1-\ell}} \bar{\tau}_i(a_i^0, a_i^1 \mid t_i, s_i, \theta).$$

By additive separability,

$$\bar{U}_i(\bar{\tau}_i, \bar{\sigma}_{-i}) = U_i^0(\tau_i^0, \sigma_{-i}^0) + U_i^1(\tau_i^1, \sigma_{-i}^1).$$

Since σ^ℓ is optimal in $(\mathcal{G}^\ell, I)$,

$$U_i^\ell(\sigma^\ell) \geq U_i^\ell(\tau_i^\ell, \sigma_{-i}^\ell), \quad \ell \in \{0, 1\}.$$

Summing over ℓ yields

$$\bar{U}_i(\bar{\sigma}) \geq \bar{U}_i(\bar{\tau}_i, \bar{\sigma}_{-i}),$$

so $\bar{\sigma}$ is a symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$.

We next prove part (2). Suppose $\bar{\sigma}$ is a symmetric Bayes Nash equilibrium of $(\bar{\mathcal{G}}, I)$, and let σ^ℓ be its component marginal. Symmetry of each σ^ℓ follows from symmetry of $\bar{\sigma}$. Fix $\ell \in \{0, 1\}$. If σ^ℓ were not a Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$, then for some player i there would exist an alternative strategy τ_i^ℓ such that

$$U_i^\ell(\tau_i^\ell, \sigma_{-i}^\ell) > U_i^\ell(\sigma^\ell).$$

Construct a feasible linked deviation $\bar{\tau}_i$ whose component- ℓ marginal is τ_i^ℓ and whose component- $(1-\ell)$ marginal is the original marginal induced by $\bar{\sigma}_i$. Let

$$\sigma_i^{1-\ell}(a_i^{1-\ell} \mid t_i, s_i, \theta) := \sum_{a_i^\ell \in A^\ell} \bar{\sigma}_i(a_i^0, a_i^1 \mid t_i, s_i, \theta),$$

where in the summand the component $a_i^{1-\ell}$ is held fixed and the component a_i^ℓ is the variable of summation. For $\bar{a}_i = (a_i^0, a_i^1)$, define

$$\bar{\tau}_i(\bar{a}_i \mid t_i, s_i, \theta) := \tau_i^\ell(a_i^\ell \mid t_i, s_i, \theta) \sigma_i^{1-\ell}(a_i^{1-\ell} \mid t_i, s_i, \theta).$$

By construction, the component- ℓ marginal of $\bar{\tau}_i$ is τ_i^ℓ , while the component- $(1 - \ell)$ marginal is $\sigma_i^{1-\ell}$. Hence, by additive separability of linked payoffs,

$$\bar{U}_i(\bar{\tau}_i, \bar{\sigma}_{-i}) = U_i^\ell(\tau_i^\ell, \sigma_{-i}^\ell) + U_i^{1-\ell}(\sigma_i^{1-\ell}, \sigma_{-i}^{1-\ell}).$$

Since

$$U_i^{1-\ell}(\sigma_i^{1-\ell}, \sigma_{-i}^{1-\ell}) = U_i^{1-\ell}(\sigma^{1-\ell})$$

and

$$U_i^\ell(\tau_i^\ell, \sigma_{-i}^\ell) > U_i^\ell(\sigma^\ell),$$

we have

$$\bar{U}_i(\bar{\tau}_i, \bar{\sigma}_{-i}) > U_i^\ell(\sigma^\ell) + U_i^{1-\ell}(\sigma^{1-\ell}) = \bar{U}_i(\bar{\sigma}),$$

contradicting optimality of $\bar{\sigma}$.

Therefore each σ^ℓ is a symmetric Bayes Nash equilibrium of $(\mathcal{G}^\ell, I)$. $\square$

A.3 Proof of Proposition 3

Proof. The observed marginal restriction implies that the observed component of the linked recommendation is pinned down: if $c_i = 0$, then $a_i^0 = E$ almost surely, and if $c_i = 2$, then $a_i^0 = N$ almost surely. Hence the only remaining degrees of freedom are the four counterfactual entry probabilities $p_{00}, p_{02}, p_{20}, p_{22}$.

First consider a low-cost firm, $c_i = 0$. In the counterfactual game, entry yields payoff $3+z$ if the opponent does not enter and payoff $1+z$ if the opponent enters. Both are strictly positive for $z \in [0, 1]$. Thus counterfactual nonentry cannot be obediently recommended to a low-cost firm, and therefore

$$p_{00} = p_{02} = 1.$$

Now consider a high-cost firm, $c_i = 2$. Obedience for the observed component, applied to high-cost firms who are recommended nonentry in the observed game, imposes restrictions through the linked recommendations. For the linked recommendation (N, E) , the unnormalized gain from deviating in the observed component from N to E is

$$\frac{1}{2}p_{20}(-1) + \frac{1}{2}p_{22}(1).$$

The obedience inequality requires this expression to be weakly negative, hence

$$p_{22} \leq p_{20}.$$

For the linked recommendation (N, N) , the corresponding unnormalized gain is

$$\frac{1}{2}(1 - p_{20})(-1) + \frac{1}{2}(1 - p_{22})(1),$$

so the obedience inequality requires

$$p_{20} \leq p_{22}.$$

Therefore

$$p_{20} = p_{22}.$$

Let the common value be p_H .

Given $p_{20} = p_{22} = p_H$, the high-cost firm's counterfactual recommendation does not change its posterior over the opponent's cost. The opponent is low cost with probability $1/2$ and high cost with probability $1/2$. If the opponent is low cost, the opponent enters for sure, and the high-cost firm's payoff from entry is $z - 1$. If the opponent is high cost, the opponent enters with probability p_H , and the high-cost firm's expected payoff from entry is

$$(1 - p_H)(1 + z) + p_H(z - 1) = 1 + z - 2p_H.$$

Thus the high-cost firm's expected gain from counterfactual entry is

$$\frac{1}{2}(z - 1) + \frac{1}{2}(1 + z - 2p_H) = z - p_H.$$

If $p_H > 0$, counterfactual entry is recommended with positive probability, so obedience requires $z - p_H \geq 0$. If $p_H < 1$, counterfactual nonentry is recommended with positive probability, so obedience requires $z - p_H \leq 0$. Thus, for $z \in (0, 1)$, both inequalities must hold and therefore $p_H = z$. At $z = 0$, any $p_H > 0$ would violate the entry-obedience inequality, so $p_H = 0$. At $z = 1$, any $p_H < 1$ would violate the nonentry-obedience inequality, so $p_H = 1$. Hence $p_H = z$ for every $z \in [0, 1]$.

It remains to verify that the rule just derived is feasible. Low-cost firms strictly prefer entry in both components. High-cost firms are obedient in the observed component because $p_{20} = p_{22}$, which makes the observed entry gain equal to zero. They are obedient in the counterfactual component because $p_H = z$, which makes the counterfactual entry gain equal to zero.

Since linked payoffs are additively separable across the observed and counterfactual components, the payoff gain from any linked action deviation is the sum of the gains from its two component deviations. Componentwise obedience therefore implies obedience against any linked action deviation. Hence the rule is feasible and is the unique feasible SBCE recommendation rule for the linked base game.

Finally, compute counterfactual producer surplus. For $c = (0, 0)$, both firms enter for sure, so total producer surplus is $2(1 + z)$. For each asymmetric cost profile $c = (0, 2)$ or $c = (2, 0)$, expected total producer surplus is

$$z \cdot 2z + (1 - z)(3 + z) = 3 - 2z + z^2.$$

For $c = (2, 2)$, each high-cost firm enters independently with probability z , so expected total producer surplus is

$$2z(1 - z)(1 + z) + z^2(2z - 2) = 2z - 2z^2.$$

Averaging over the four equally likely cost profiles gives

$$W(\mu^{\text{cf}}) = \frac{1}{4} \left[(2 + 2z) + (3 - 2z + z^2) + (3 - 2z + z^2) + (2z - 2z^2) \right] = 2$$

for the counterfactual distribution induced by the unique feasible SBCE recommendation rule of the linked base game. Therefore

$$\mathcal{W}^{\text{SBCE}}(z) = \{2\}$$

for every $z \in [0, 1]$. □

B Primitive representation of the SBCE feasible set

This appendix gives a primitive representation of the feasible set $\mathcal{F}(\mu^{\text{obs}})$ used in Theorem 2. The representation makes explicit how the projection characterization is implemented through a symmetric independent recommendation rule for the linked base game.

For a given observed component-game distribution

$$\mu^{\text{obs}} \in \Delta(\mathcal{A}^0 \times T^N \times \Theta),$$

Theorem 2 states that

$$\mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}}) = \text{proj}_{\mathcal{A}^1 \times T^N \times \Theta} \mathcal{F}(\mu^{\text{obs}}).$$

The remainder of this appendix unpacks the feasible set $\mathcal{F}(\mu^{\text{obs}})$ in primitive form.

B.1 Primitive recommendation rule and induced distribution

Fix the two component games $\ell \in \{0, 1\}$, the common private type space T , and the common public state space Θ . Also fix the type-state marginal distribution

$$\pi^{T\Theta} \in \Delta(T^N \times \Theta).$$

These objects define the linked base game

$$\bar{\mathcal{E}} = (\bar{\mathcal{G}}, \pi^{T\Theta}).$$

For each player i , let

$$\bar{A}_i := A^0 \times A^1$$

denote the linked action space, and let

$$\bar{\mathcal{A}} := \prod_{i=1}^N \bar{A}_i$$

denote the linked action-profile space. Write $\bar{\mathcal{A}}_{-i} := \prod_{j \neq i} \bar{A}_j$ for the opponents' linked action-profile space. A generic linked action is written

$$\bar{a}_i = (a_i^0, a_i^1) \in \bar{A}_i.$$

For each private type profile $t \in T^N$, let

$$n_{-i}(t_{-i}) \in \mathcal{H}$$

denote the anonymous summary of opponents' private types, where

$$\mathcal{H} := \left\{ m \in \mathbb{Z}_+^{|T|} : \sum_{\tau \in T} m^\tau = N - 1 \right\}.$$

The primitive SBCE object is a symmetric independent recommendation rule for the linked base game, represented by a stochastic kernel

$$\bar{\nu}(\cdot \mid t_i, m, \theta) \in \Delta(\bar{A}_i), \quad (t_i, m, \theta) \in T \times \mathcal{H} \times \Theta.$$

Equivalently, for each conditioning object

$$(t_i, m, \theta) \in T \times \mathcal{H} \times \Theta,$$

the primitive variables are

$$\{\bar{\nu}(\bar{a}_i \mid t_i, m, \theta) : \bar{a}_i \in \bar{A}_i\}.$$

Given $\bar{\nu}$, define the induced distribution over $\bar{\mathcal{A}} \times T^N \times \Theta$ by

$$\mu_{\bar{\nu}}(\bar{a}, t, \theta) = \pi^{T\Theta}(t, \theta) \prod_{i=1}^N \bar{\nu}(\bar{a}_i \mid t_i, n_{-i}(t_{-i}), \theta), \quad (\bar{a}, t, \theta) \in \bar{\mathcal{A}} \times T^N \times \Theta.$$

Thus the full distribution over $\bar{\mathcal{A}} \times T^N \times \Theta$ is not chosen directly. It is generated from the lower-dimensional primitive object $\bar{\nu}$ through the product structure of the symmetric independent recommendation rule.

B.2 Primitive constraints

For every $(t_i, m, \theta) \in T \times \mathcal{H} \times \Theta$, the recommendation rule must satisfy the simplex constraints

$$\bar{\nu}(\bar{a}_i \mid t_i, m, \theta) \geq 0 \quad \forall \bar{a}_i \in \bar{A}_i, \quad \sum_{\bar{a}_i \in \bar{A}_i} \bar{\nu}(\bar{a}_i \mid t_i, m, \theta) = 1.$$

Bayes consistency with respect to the given type-state marginal is built into the construction. Indeed,

$$\sum_{\bar{a} \in \bar{\mathcal{A}}} \mu_{\bar{\nu}}(\bar{a}, t, \theta) = \pi^{T\Theta}(t, \theta) \quad \forall (t, \theta) \in T^N \times \Theta.$$

The observed component must reproduce the given observed object:

$$\text{proj}_{\mathcal{A}^0 \times T^N \times \Theta}(\mu_{\bar{\nu}}) = \mu^{\text{obs}}.$$

Equivalently, for every

$$(a^0, t, \theta) \in \mathcal{A}^0 \times T^N \times \Theta,$$

one requires

$$\sum_{a^1 \in \mathcal{A}^1} \mu_{\bar{\nu}}(a^0, a^1, t, \theta) = \mu^{\text{obs}}(a^0, t, \theta).$$

Substituting the induced distribution yields

$$\sum_{a^1 \in \mathcal{A}^1} \pi^{T\Theta}(t, \theta) \prod_{i=1}^N \bar{\nu}((a_i^0, a_i^1) \mid t_i, n_{-i}(t_{-i}), \theta) = \mu^{\text{obs}}(a^0, t, \theta),$$

which is polynomial in the primitive variables.

Next, recall that the linked payoff of player i is

$$\bar{u}_i(\bar{a}, t, \theta) = u_i^0(a^0, t, \theta) + u_i^1(a^1, t, \theta), \quad \bar{a} = (a^0, a^1).$$

The recommendation rule must satisfy obedience for the linked base game. For every player i , every private type $t_i \in T$, every public state $\theta \in \Theta$, every recommended linked action $\bar{a}_i \in \bar{A}_i$, and every deviation linked action $\bar{a}'_i \in \bar{A}_i$, one requires

$$\sum_{t_{-i} \in T^{N-1}} \sum_{\bar{a}_{-i} \in \bar{A}_{-i}} \mu_{\bar{\nu}}(\bar{a}_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) \left[\bar{u}_i(\bar{a}_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) - \bar{u}_i(\bar{a}'_i, \bar{a}_{-i}, t_i, t_{-i}, \theta) \right] \geq 0.$$

When the conditioning event has zero probability, the unnormalized inequality is vacuous. After substitution, these are polynomial inequalities in the primitive variables.

B.3 Feasible set, projection, and extremal values

The feasible set appearing in Theorem 2 is the set of distributions $\mu_{\bar{\nu}}$ over $\bar{\mathcal{A}} \times T^N \times \Theta$ generated by primitive rules $\bar{\nu}$ that satisfy the simplex constraints, observed marginal matching, and obedience for the linked base game:

$$\mathcal{F}(\mu^{\text{obs}}) := \{ \mu_{\bar{\nu}} \in \Delta(\bar{\mathcal{A}} \times T^N \times \Theta) : \bar{\nu} \text{ satisfies the primitive SBCE constraints above} \}.$$

For any feasible $\bar{\nu}$, the induced counterfactual marginal is

$$\mu_{\bar{\nu}}^{\text{cf}}(a^1, t, \theta) = \sum_{a^0 \in \mathcal{A}^0} \mu_{\bar{\nu}}(a^0, a^1, t, \theta).$$

Hence

$$\mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}}) = \{ \mu_{\bar{\nu}}^{\text{cf}} : \bar{\nu} \text{ satisfies the primitive SBCE constraints above} \}.$$

Let W be a counterfactual statistic defined on

$$\Delta(\mathcal{A}^1 \times T^N \times \Theta).$$

The associated SBCE statistic prediction set is

$$\mathcal{W}^{\text{SBCE}}(\mu^{\text{obs}}) := \{ W(\mu_{\bar{\nu}}^{\text{cf}}) : \bar{\nu} \text{ satisfies the primitive SBCE constraints above} \}.$$

The corresponding lower and upper bounds are

$$\underline{W}^{\text{SBCE}}(\mu^{\text{obs}}) := \inf_{\mu^{\text{cf}} \in \mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}})} W(\mu^{\text{cf}}), \quad \overline{W}^{\text{SBCE}}(\mu^{\text{obs}}) := \sup_{\mu^{\text{cf}} \in \mathcal{C}^{\text{SBCE}}(\mu^{\text{obs}})} W(\mu^{\text{cf}}).$$

These lower and upper bounds do not by themselves imply that every value between them is attained. They are the infimum and supremum of the statistic image set, not a statement that the image set is an interval.

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